

Section : Physics

- Q.1** The elements which may be considered for an ideal inductor as a solenoid,
 (A) which is made of a thick wire
 (B) which has negligible resistance
 (C) which has a sufficiently large inductance
 (D) which is made of a thin wire
 Choose the correct answer from the options given below:
 1. (A), (B) and (C) only
 2. (B) and (C) only
 3. (A), (B), (C) and (D)
 4. (B), (C) and (D) only

Options 1. 1
 2. 2
 3. 3
 4. 4

- Q.2** Identify the correct statements.
 (A) Electric charge is quantized.
 (B) Electric charge on any object is always an integral multiple of e .
 (C) The coulomb force is weaker than the gravitational force.
 (D) The coulomb force is a conservative force.
 Choose the correct answer from the options given below:
 1. (A), (B) and (C) only
 2. (A) and (B) only
 3. (B), (C) and (D) only
 4. (A), (B) and (D) only

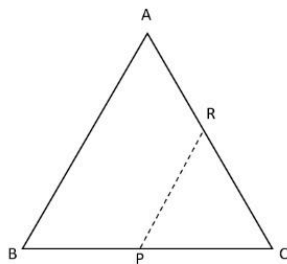
Options 1. 1
 2. 2
 3. 3
 4. 4

- Q.3** A long straight wire is carrying a current of 20 A. An electron is moving with a speed of 10^7 m/s parallel to the wire.
 The force experienced by the electron when it is 2 cm away from the wire is:

1. $3.2\pi \times 10^{-16}$ N
2. 3.2×10^{-4} N
3. $\frac{3.2}{\pi} \times 10^{-10}$ N
4. 3.2×10^{-16} N

Options 1. 1
 2. 2
 3. 3
 4. 4

- Q.4** Three charges, each equal to $+q$, are placed at the corners of an equilateral triangle ABC of side 'a' as shown in the figure. P and R are the mid-points of BC and CA. The work done in taking a charge Q from point P to R is:



1. $\frac{3qQ}{4\pi\epsilon_0 a}$
2. $\frac{3qQ}{12\pi\epsilon_0 a}$
3. $\frac{qQ}{4\pi\epsilon_0 a}$
4. Zero

Options 1. 1
 2. 2
 3. 3
 4. 4

- Q.5** Three concentric metallic spherical shells of radii R , $2R$, and $3R$ are given charges Q_1 , Q_2 and Q_3 , respectively.
 It is found that the surface charge densities on the outer surfaces of the shells are equal. Then, the ratio of the charges given to the shells $Q_1 : Q_2 : Q_3$ is:

1. 1 : 2 : 3
2. 1 : 4 : 9
3. 1 : 3 : 5

4. 1 : 8 : 18

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.6 In the single slit experiment of diffraction, the aperture of the slit is 2 mm.

If a monochromatic light of wavelength 600 nm is incident normally on the slit, the distance between 2nd order minima and the 4th order maxima on the screen placed 2√3 m away from the slit will be:

- 1. 0.775 mm
- 2. 1 mm
- 3. 3 mm
- 4. 1.5 mm

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.7 The magnetic flux linked with a coil at any instant t is given by $\Phi = (4t^3 - 3t^2 + 2t + 5)$ Wb. The emf induced in the coil at $t = 3$ s is

- 1. 128 V
- 2. - 128 V
- 3. 133 V
- 4. - 92 V

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.8 A lamp rating 8 V, 800 W is connected to a 100 V, 50 Hz a. c. source. The inductance of the choke required in the circuit is: (Take $\pi = 3$)

- 1. 2 mH
- 2. 20 mH
- 3. 3.3 mH
- 4. 10 mH

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.9 A potential difference of 6 V is applied across a conductor of length 12 cm.

If the mobility of electrons is $5.6 \times 10^{-6} \text{ m}^2/\text{V}\cdot\text{s}$, their drift velocity will be:

- 1. $2.8 \times 10^{-3} \text{ m/s}$
- 2. $5.6 \times 10^{-3} \text{ m/s}$
- 3. $2.8 \times 10^{-4} \text{ m/s}$
- 4. $0.56 \times 10^{-4} \text{ m/s}$

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.10 The current and power dissipation in an inductor coil are 2 A and 200 W, respectively.

When connected to an alternating source of 220 V, 50 Hz. The resistance and inductance of the coil are (take $\pi^2 = 10$):

- 1. 50Ω and 0.511 H
- 2. 50Ω and 0.312 H
- 3. 100Ω and 0.511 H
- 4. 100Ω and 0.312 H

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.11 Arrange the following in the order of increasing resistivity (ρ)

- (A) Copper
- (B) Silver
- (C) Glass
- (D) Carbon (graphite)
- (E) Germanium

Choose the correct answer from the options given below:

- 1. (B), (A), (D), (E), (C)
- 2. (B), (D), (A), (E), (C)
- 3. (B), (A), (D), (C), (E)
- 4. (D), (B), (E), (C), (A)

Options 1. 1

2. 2
3. 3
4. 4

Q.12 Match List-I with List-II

List-I

Uses of electromagnetic waves

- (A) Aircraft navigation
- (B) Cellular phones
- (C) Remote switches for TV sets
- (D) LASIK eye surgery

List-II

Type of electromagnetic waves

- (I) Infrared waves
- (II) Microwaves
- (III) Ultraviolet rays
- (IV) Radio waves

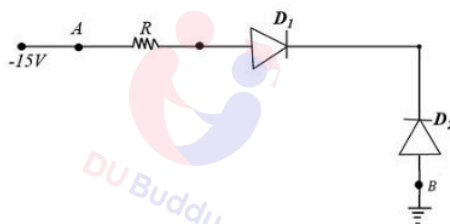
Choose the correct answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
2. (A) - (IV), (B) - (I), (C) - (II), (D) - (III)
3. (A) - (II), (B) - (IV), (C) - (I), (D) - (III)
4. (A) - (I), (B) - (IV), (C) - (II), (D) - (III)

- Options**
1. 1
 2. 2
 3. 3
 4. 4

Q.13

In the figure, assuming the diodes to be ideal:-

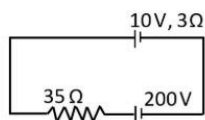


1. D_2 is forward biased and D_1 is reverse biased and hence no current flows from B to A and vice-versa
2. D_1 is forward biased and D_2 is reverse biased and hence current flows from A to B
3. D_1 and D_2 are both forward biased and hence current flows from A to B
4. D_1 and D_2 are both reverse biased and hence no current flows from A to B and vice-versa

- Options**
1. 1
 2. 2
 3. 3
 4. 4

Q.14

A 10 V battery having $3\ \Omega$ internal resistance is connected with a 200 V battery and a $35\ \Omega$ resistance as shown in the figure. The terminal potential drop at 10 V battery will be:



1. 18 V
2. 25 V
3. 15 V
4. 9 V

- Options**
1. 1
 2. 2
 3. 3
 4. 4

Q.15 A yellow LED of power 0.2 W emits light uniformly around it. The amplitude of the electric field of light at a distance of 2 m from the LED is:

1. 2.45 V/m
2. 5.48 V/m
3. 4.9 V/m
4. 1.73 V/m

- Options**
1. 1
 2. 2
 3. 3
 4. 4

Q.16 The wavelength of second line of Balmer series in terms of Rydberg constant R is

1. $\frac{36}{5R}$

2. $\frac{16}{5R}$
3. $\frac{16}{3R}$
4. $\frac{36}{3R}$

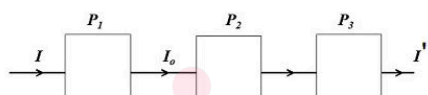
Options 1. 1
2. 2
3. 3
4. 4

Q.17 The distance between a luminous object and the screen is 60 cm. A convex lens placed in between produces a three times magnified image on the screen. The focal length of the convex lens is:

1. 11.25 cm
2. 10.5 cm
3. 15.0 cm
4. 10.25 cm

Options 1. 1
2. 2
3. 3
4. 4

Q.18



P_1, P_2, P_3 are three Polaroid sheets, P_1 and P_3 are two crossed polaroids. What should be the angle between Polaroids P_1 and P_2 , so that the value of $I' = I_0/4$

1. $\theta = \pi/2$
2. $\theta = \pi/4$
3. $\theta = \pi/6$
4. $\theta = \pi/8$

Options 1. 1
2. 2
3. 3
4. 4

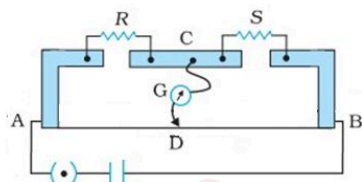
Q.19 A p-type semiconductor can be obtained by doping of:

1. Arsenic to pure silicon
2. Indium to pure silicon
3. Antimony to pure germanium
4. Phosphorous to pure germanium

Options 1. 1
2. 2
3. 3
4. 4

Q.20

In a meter bridge, the null point is found at a distance of 60 cm from end A. If a $5\ \Omega$ resistor is now connected in series with S, the null point occurs at 50 cm. The values of R and S are, respectively



1. $10\ \Omega$ and $15\ \Omega$
2. $15\ \Omega$ and $10\ \Omega$
3. $9\ \Omega$ and $6\ \Omega$
4. $12\ \Omega$ and $8\ \Omega$

Options 1. 1
2. 2
3. 3
4. 4

Q.21 A solenoid of length 30 cm and 3000 turns carries a current of 10 A. The value of the magnetic field ($\vec{B}$) at the centre of solenoid is:
(Given $\mu_0 = 4\pi \times 10^{-7}\ \text{Wb/A-m}$)

1. $0.13\ \text{Wb/m}^2$
2. $1.256\ \text{Wb/m}^2$
3. $12.56\ \text{Wb/m}^2$
4. $0.628\ \text{Wb/m}^2$

Options 1. 1

2. 2
3. 3
4. 4

Q.22 Two particles x and y having equal charge, after being accelerated through the same potential difference, enter a region of a uniform magnetic field. They describe circular paths of radii R_1 and R_2 , respectively. The ratio of the mass of x to that of y is:

1. $\sqrt{\frac{R_1}{R_2}}$
2. $\frac{R_1}{R_2}$
3. $\left(\frac{R_1}{R_2}\right)^2$
4. $R_1 R_2$

Options 1. 1

2. 2
3. 3
4. 4

Q.23 A moving coil galvanometer has 100 equal divisions. Its current sensitivity is 20 divisions per mA and voltage sensitivity is 4 divisions per mV. To convert it into a voltmeter in which each division reads 1 V, the resistance needed to be connected in series with the coil will be :

1. 9995 Ω
2. 19995 Ω
3. 995 Ω
4. 4995 Ω

Options 1. 1

2. 2
3. 3
4. 4

Q.24 Identify the correct statements:

- (A) A magnetic field exerts force on a moving charge and not on a stationary charge.
- (B) A moving charge produces a magnetic field.
- (C) The magnetic field at a point on the axis of a current element is zero.
- (D) Magnetic field $\vec{B}$ is tangential to a magnetic field line.

Choose the correct answer from the options given below:

1. (A), (B) (C) and (D)
2. (B), (C) and (D) only
3. (A), (B) and (D) only
4. (A), (B) and (C) only

Options 1. 1

2. 2
3. 3
4. 4

Q.25 The refractive index of water is $\left(\frac{4}{3}\right)$ and that of glass is $\left(\frac{3}{2}\right)$.

If the speed of light in glass is 2×10^8 m/s, the speed of light in water will be:

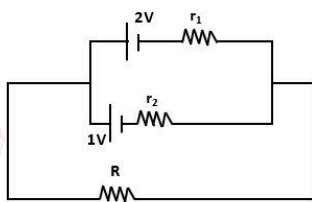
1. $\frac{8}{3} \times 10^8$ m/s
2. 4×10^8 m/s
3. $\left(\frac{9}{4}\right) \times 10^8$ m/s
4. 1×10^8 m/s

Options 1. 1

2. 2
3. 3
4. 4

Q.26

Two cells with internal resistances r_1 and r_2 are connected across the load resistance R as shown in the figure. The values of r_1 , r_2 and R are $(1/2) \Omega$, $(1/3) \Omega$ and $(9/5) \Omega$, respectively. The current through R is:



1. $7/5$ A
2. $7/15$ A
3. $15/7$ A
4. $7/10$ A

Options 1. 1

2. 2
3. 3
4. 4

Q.27 The number of photons emitted per minute by a 25 W source of monochromatic light of wavelength 6000 Å is approximately:
(take $h = 6.6 \times 10^{-34}$ Js)

1. 2.5×10^{21}
2. 4.5×10^{21}
3. 4.5×10^{21}
4. 5.5×10^{21}

Options 1. 1

2. 2
3. 3
4. 4

Q.28 The stopping potential of photo-electrons depends upon:

1. Frequency of incident light only
2. material of the cathode only
3. both the frequency of the incident light and the material of the cathode.
4. intensity of the incident light.

Options 1. 1

2. 2
3. 3
4. 4

Q.29 A thin convex lens of refractive index 1.5 has focal length f_1 . When immersed in a liquid of refractive index $5/3$, its focal length is found to be f_2 . The relationship between f_1 and f_2 is:

1. $f_2 = -5f_1$
2. $f_2 = -f_1/5$
3. $f_1 = -5f_2$
4. $f = -3f_2$

Options 1. 1

2. 2
3. 3
4. 4

Q.30 Identify the correct statements.

- (A) Neutrons penetrate the nucleus more readily as compared to protons.
- (B) It is easier to release an orbital electron, but quite difficult to release a nucleon.
- (C) Nuclear density is not the same for all nuclei.
- (D) The radius of the nucleus is directly proportional to the cube root of the mass number.

Choose the correct answer from the options given below:

1. (A) and (B) only
2. (A), (B) and (D) only
3. (B), (C) and (D) only
4. (A) and (C) only

Options 1. 1

2. 2
3. 3
4. 4

Q.31 The capacitance of a capacitor becomes $\frac{9}{7}$ times of its original value when a dielectric slab of thickness $t = \frac{d}{4}$ is inserted between the plates, where d

is the separation between the plates. The dielectric constant of the slab is

1. 7
2. $\frac{7}{15}$
3. 9
4. $\frac{15}{7}$

Options 1. 1

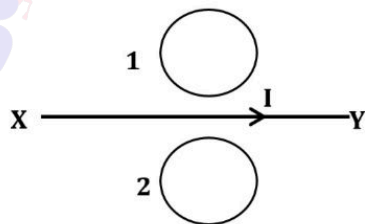
2. 2

3. 3

4. 4

Q.32

In the given arrangement, the current in the wire XY is increasing steadily. The direction of induced current in the metal rings 1 and 2 will be, respectively.



1. anticlockwise, clockwise

2. both clockwise

3. both anticlockwise

4. clockwise, anticlockwise

Options 1. 1

2. 2

3. 3

4. 4

Q.33

A wire has a resistance of $3\ \Omega$ at 20°C and $5\ \Omega$ at 100°C . The temperature coefficient of resistance of the wire is:

1. $0.00833\ ^\circ\text{C}^{-1}$

2. $0.0125\ ^\circ\text{C}^{-1}$

3. $0.0025\ ^\circ\text{C}^{-1}$

4. $0.0012\ ^\circ\text{C}^{-1}$

Options 1. 1

2. 2

3. 3

4. 4

Q.34 The wavelength of the de-Broglie wave associated with an electron accelerated through 150 V is:

1. 1 Å

2. 2 Å

3. 2.7 Å

4. 12.7 Å

Options 1. 1

2. 2

3. 3

4. 4

Q.35

A bar magnet of magnetic moment M is bent into a semicircular form. Its new magnetic moment will be:

1. M

2. $\frac{M}{\pi}$

3. $\frac{2M}{\pi}$

4. $\frac{3M}{\pi}$

Options 1. 1

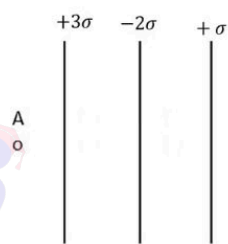
2. 2

3. 3

4. 4

Q.36

In the figure, there are three infinitely long thin sheets having surface charge density $+3\sigma$, -2σ and $+\sigma$ respectively. Determine magnitude and direction of the electric field at point A.



1. $\frac{\sigma}{\epsilon_0}$ towards right
2. $\frac{\sigma}{2\epsilon_0}$ towards left
3. $\frac{2\sigma}{\epsilon_0}$ towards right
4. $\frac{\sigma}{\epsilon_0}$ towards left

Options 1. 1

2. 2

3. 3

4. 4

Q.37 Keeping the voltage of the charging source constant, the separation between the plates of the capacitor is decreased by 10%. The percentage change in the energy stored in the capacitor will be:

1. 11.1%, Increase
2. 11.1%, Decrease
3. 15.1%, Increase
4. 15.1%, Decrease

Options 1. 1

2. 2

3. 3

4. 4

Q.38 When a p-n junction is reverse biased:

1. Width of depletion region decreases
2. The drift of electrons and holes stops.
3. Potential barrier decreases
4. The diffusion of electrons and holes stops

Options 1. 1

2. 2

3. 3

4. 4

Q.39 In a plane electromagnetic wave, the electric field oscillates sinusoidally at a frequency of 3×10^{10} Hz and its amplitude is 30 V/m. The amplitude of the oscillating magnetic field is:

1. 1×10^{-7} T
2. 1.5×10^{-7} T
3. 1.5×10^{-8} T
4. 1×10^{-8} T

Options 1. 1

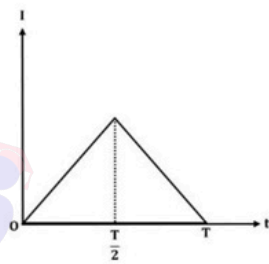
2. 2

3. 3

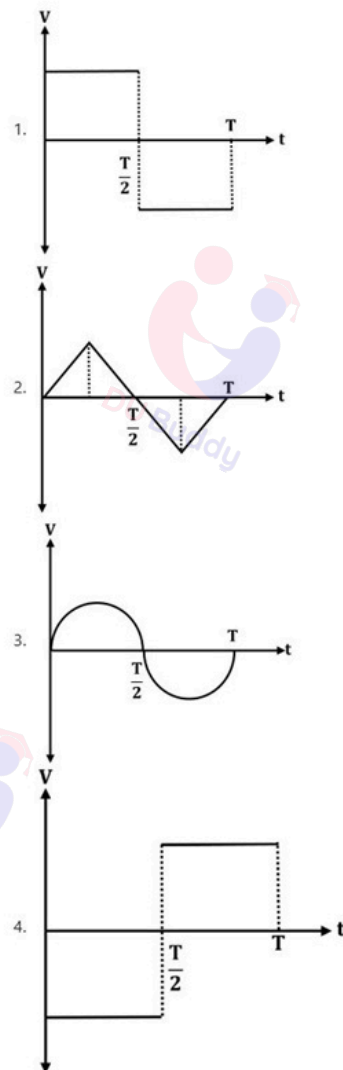
4. 4

Q.40

The graph shows variation of current (I) with time (t) in an inductance coil.



The graph that depicts the correct variation of induced emf produced in the coil with time is:



Options

1. 1
2. 2
3. 3
4. 4

Q.41

Eight charged water droplets, each of radius 1 mm and carrying a charge 5×10^{-10} C, coalesce to form a single drop. The potential of the bigger drop is.

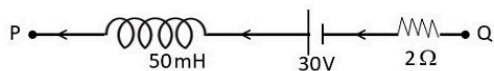
1. 36000 V
2. 18000 V
3. 15000 V
4. 30000 V

Options

1. 1
2. 2
3. 3
4. 4

Q.42

At some instant, the current in the circuit shown below is 1 A, and it is decreasing at a rate of 10^2 A/s. The value of the potential difference ($V_P - V_Q$) at that instant is:



1. 33 V
2. 120 V
3. 23 V
4. 66 V

Options

1. 1
2. 2
3. 3
4. 4

Q.43 To observe diffraction, the size of an obstacle

1. Should be of the same order as wavelength
2. Should be much larger than the wavelength
3. has no relation with wavelength
4. should be exactly half of the wavelength

Options

1. 1
2. 2
3. 3
4. 4

Q.44 In a double slit experiment using light of wavelength 450 nm, the angular width of a fringe formed on a distant screen is 0.15° . The spacing between the two slits is

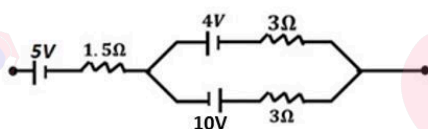
1. 1.72×10^{-4} m
2. 2.54×10^{-4} m
3. 3.54×10^{-4} m
4. 4.172×10^{-4} m

Options

1. 1
2. 2
3. 3
4. 4

Q.45

Three cells with internal resistance are connected as shown in the figure. The equivalent emf of the combination of three cells is



1. 2 V
2. 3 V
3. 5 V
4. 10 V

Options

1. 1
2. 2
3. 3
4. 4

Q.46 Ionization potential of a hydrogen atom is 13.6 V. Hydrogen atom in the ground state is excited by monochromatic radiation of photon energy 12.75 eV. The number of spectral lines emitted by hydrogen atom will be

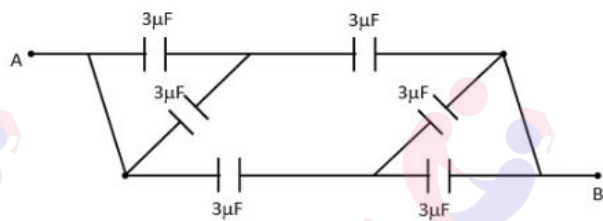
1. 3
2. 4
3. 6
4. 5

Options

1. 1
2. 2
3. 3
4. 4

Q.47

Find the equivalent capacitance between the points A and B.



1. $4 \mu\text{F}$
2. $3 \mu\text{F}$
3. $18 \mu\text{F}$
4. $6 \mu\text{F}$

Options

1. 1
2. 2
3. 3
4. 4

Q.48 An ammeter of resistance 0.54Ω reads current upto 1 A. The value of the shunt required to increase the range to 10 A is

1. 0.09Ω
2. 0.05Ω
3. 0.06Ω
4. 0.07Ω

Options

1. 1
2. 2
3. 3
4. 4

Q.49 A 12 V battery is connected to a 6Ω , 10 H coil, through a closed switch. The switch is suddenly opened in 1 ms. The average induced emf across the coil will be-

1. 10,000V
2. 20,000V
3. 15,000V
4. 30,000V

Options

1. 1
2. 2
3. 3
4. 4

Q.50

In the nuclear decay given below, the particles emitted in the sequence are:



1. α, β, γ
2. γ, β, α
3. β, γ, α
4. β, α, γ

Options

1. 1
2. 2
3. 3
4. 4